

Rebalancing a Concentrated-Liquidity Position: Policy, Costs, Evidence

Marco Amendola · July 2026 · Two **disjoint** 90-day regimes of real ETH-USD hourly data · 400 synthetic paths (OU / GBM) · USD- and ETH-denominated baselines · Measured pool fee anchor · Code, validation suite and research log ship with this document

Scope

This study answers “**how should a Uniswap-v3-style LP position be rebalanced?**” with measured evidence across two disjoint market regimes and a synthetic mechanism test. It is the decision layer — whether and how to rebalance, with explicit costs. A production system adds measured position-level fee accrual, tick-level data and broader validation on top; the architecture is built to take them. The theoretical anchor is loss-versus-rebalancing (Milionis, Moallemi, Roughgarden & Zhang, 2022): this work operationalises that cost into a per-policy, per-regime break-even that a desk can act on.

Method

Exact v3 position mathematics (no linearised IL approximations), validated by a nine-test suite (value continuity at bounds, out-of-range composition, reduction to the v2 square-root law) and a reproduction gate: before any new run, the engine re-derived every published v2 number from archived inputs (exact) and from re-fetched data (0.00pp deviation on all rows; mode logged per window). Six policies were run over two real 90-day hourly ETH-USD windows — **A: trending** (9 Apr–8 Jul 2026, net −18.5%, σ 55% ann., daily path-efficiency 0.120) and **B': oscillatory** (6 Aug–3 Nov 2024, net −2.3%, σ 56% ann., daily path-efficiency 0.004) — against both a 50/50 HODL baseline (USD terms) and a 100%-ETH hold (ETH terms). B' was selected by a pre-registered mechanical scan (end $\leq$ 8 Apr 2026 for full disjointness from A; $|\text{net}| \leq 5\%$; $\sigma \in [40, 70]\%$; minimum daily path-efficiency; top-ranked window taken, no override). An erratum applies to v2: its windows A and B shared 53 of 90 days ($\approx 59\%$ overlap), overstating evidential independence; B' repairs it and reproduces v2-B's conclusions on independent data. Every rebalance is charged 5+5 bps on swapped value plus gas. Fee income is **inverted, not assumed**: the primary output is the fee APR each policy required to match HODL. **Convention**: break-even APR is quoted on deposited capital over the full window; time-in-range dilution applies symmetrically to required and realized accrual, and in-range equivalents ($\text{BE} \div \text{TIR}$) are reported for static ranges. Predictions were pre-registered before every run: two of four Run-002 predictions failed and produced the main finding; Run 003 (disjoint window) went 5/5; Run 004 (synthetic) went 4/4 — all logged.

Results — Window A: trending (ETH −18.5%)

Policy	vs HODL (USD)	vs 100% ETH (ETH terms)	Rebal.	Event costs	Time in range	Break-even fee APR
Full range	−0.47%	+10.78%	0	\$0	100%	1.9%
Static $\pm 15\%$	−6.31%	+3.62%	0	\$0	61%	25.6%
Static $\pm 5\%$	−8.25%	+1.23%	0	\$0	24%	33.5%
Threshold re-center $\pm 15\%$	−8.88%	+0.47%	2	\$91	100%	36.1%
Threshold re-center $\pm 5\%$	−27.45%	−22.32%	21	\$888	100%	111.6%
Periodic weekly $\pm 10\%$	−11.54%	−2.81%	12	\$316	96%	46.9%

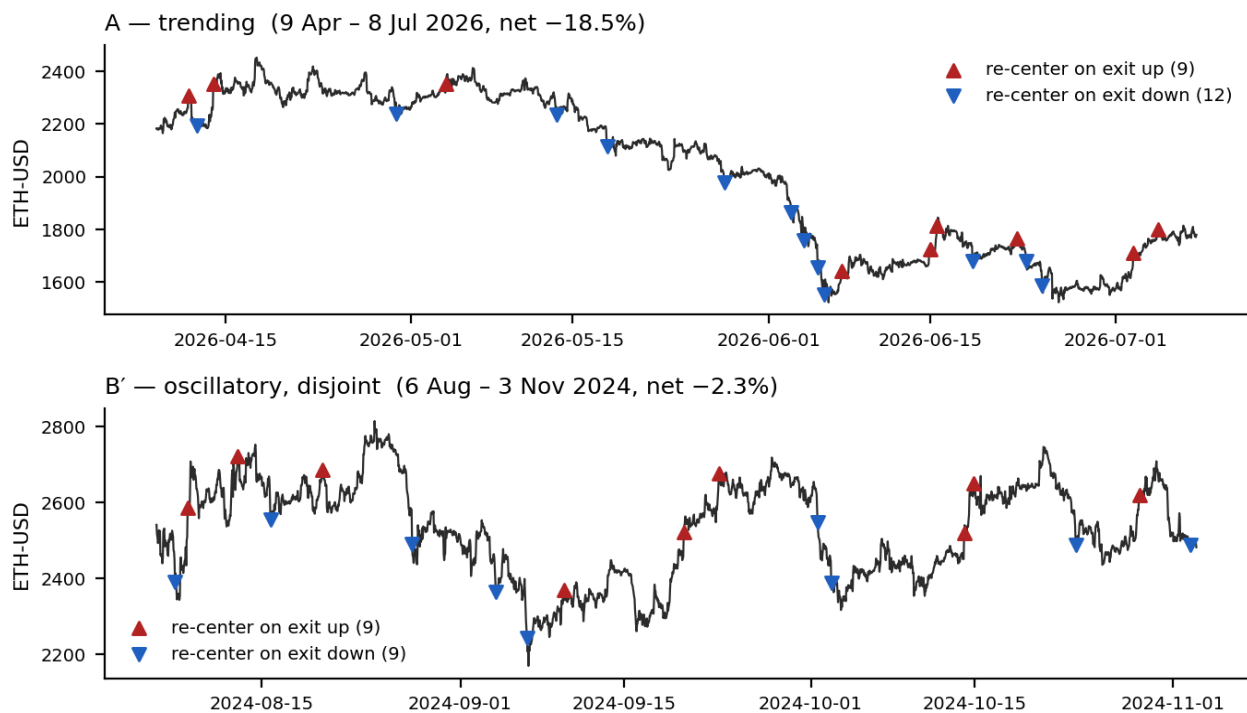
Deposit \$100,000; values pre-fee-income, net of rebalancing costs. ETH-terms column: position value in ETH at horizon vs holding 100% ETH — in a drawdown, every no-rebalance LP policy beats holding the asset in its own terms. Denomination is a first-order design choice, not a reporting detail. In-range-equivalent break-evens for the statics ($\text{BE} \div \text{TIR}$): $\pm 15\% \rightarrow 42.0\%$, $\pm 5\% \rightarrow \approx 140\%$.

Results — Window B': oscillatory, disjoint (ETH −2.3%, σ matched)

Policy	vs HODL (USD)	vs 100% ETH (ETH terms)	Rebal.	Event costs	Time in range	Break-even fee APR
Full range	−0.01%	+1.19%	0	\$0	100%	0.0%
Static ±15%	−0.10%	+1.09%	0	\$0	100%	0.4%
Static ±5%	−0.29%	+0.90%	0	\$0	67%	1.2%
Threshold re-center ±15%	−6.90%	−5.86%	2	\$98	100%	28.3%
Threshold re-center ±5%	−26.34%	−25.78%	18	\$811	100%	108.0%
Periodic weekly ±10%	−11.41%	−10.49%	12	\$294	91%	46.8%

Same conventions as window A. In-range-equivalent break-even for static ±5%: $1.2\% \div 0.67 \approx 1.8\%$. Threshold ±5% damage decomposition on B': \$811 explicit costs of a \$26,344 shortfall — 3.1% costs, 96.9% crystallised inventory drift — the same signature as window A. The five largest B' re-centers each swap $\approx$ half the position at the extreme that forced the exit (max event cost \$51.75).

Threshold ±5% re-centering: every trigger buys the extreme that forced it



Threshold ±5% re-center events over both windows. Upward triangles: re-centers triggered by an exit above the range (the policy buys back in at a local high); downward: exits below (it sells at a local low). The clustering at extremes is the mechanism, visible by eye in both regimes.

Synthetic mechanism test — OU vs GBM (Run 004, pre-registered)

Two competing mechanism labels fit the real-window evidence: the policy is short *mean reversion* at its trigger scale (v2's reading), or short *realized variance* regardless of path structure. The discriminating test: run it on pure mean-reverting paths AND on zero-drift diffusion of identical volatility — if the first label is right, it must lose systematically more where reversion exists. 200 Ornstein–Uhlenbeck paths (log-price, stationary sd 7%, half-life 9.2d, σ 52% ann.) and 200 zero-drift GBM paths (same σ), 2,160 hourly steps each, seed 42, identical costs. These are the study's first interval estimates.

Process	Policy	Break-even APR, median	5–95th percentile	Rebalances (median)
OU — pure mean reversion	Full range	0.1%	[0.0%, 1.0%]	0
	Static $\pm 5\%$	4.1%	[0.0%, 23.6%]	0
	Threshold $\pm 5\%$	116.5%	[89.7%, 141.3%]	24
GBM — zero drift	Full range	1.4%	[0.0%, 9.1%]	0
	Static $\pm 5\%$	29.3%	[1.0%, 77.7%]	0
	Threshold $\pm 5\%$	115.9%	[86.0%, 161.4%]	24

Threshold $\pm 5\%$ loses to static $\pm 5\%$ in 200 of 200 OU paths (paired); its 5th-percentile break-even on its friendliest process is 89.7% — still unpayable. The discriminating comparison (Run 004b, paired, same seed): OU median 115.8% vs GBM 119.0%, OU worse in only 43% of paired paths — **no** systematic mean-reversion penalty. A σ -sweep (40–80% ann.) moves the median from 71% to 219% near-identically on both processes. Damage tracks realized variance, not path structure. Both real windows (111.6% at σ 55%, 108.0% at σ 56%) sit inside the σ -matched synthetic 5–95 bands; higher- σ windows scale accordingly. The regime-dependent object is the static tight range: median 4.1% on OU vs 29.3% on GBM.

The finding

The v2 pre-registered hypothesis was that exit-triggered re-centering fails in trends and recovers in oscillation. The data rejected it, and v3 upgrades the rejection from observed to structural — while correcting v2's mechanism label, which the synthetic control falsified. Threshold $\pm 5\%$ is the worst policy in **both** disjoint real regimes (break-even 111.6% trending, 108.0% oscillatory; explicit costs 3–4% of the damage, crystallised inventory drift 96–97%) and in **400 of 400** synthetic paths — with statistically indistinguishable damage on pure mean reversion and on zero-drift diffusion. Mechanism: re-centering on every exit holds the position permanently at maximum curvature — always centred, always 50/50 — so it pays the concentrated-liquidity analogue of loss-versus-rebalancing continuously, at a rate set by realized variance at the trigger scale. The policy is **short realized variance**, a discrete LVR maximiser (Milionis et al., 2022), invariant to path structure and macro regime: near-identical σ explains why the two real windows price within 4pp of each other despite opposite regimes. The regime-dependent object is the static tight range: its break-even collapses from 33.5% (trending) to 1.2% (disjoint oscillatory); the order-of-magnitude difference lives in the range decision, not the re-centering decision.

Fee adequacy — measured anchor

Pool-level fee APY (fees $\div$ TVL) on the Uniswap v3 ETH/USDC 0.05% mainnet pool (0x88e6...5440), measured from the DeFiLlama daily series: **window A mean 14.9%** (median 12.0%, avg TVL \$95M); **window B' mean 17.5%** (median 16.3%, avg TVL \$151M). A Dune query computing the same metric from dex.trades and daily pool balances is published as query 7923963 (execution pending account credits — stated, not hidden). Against these measurements: threshold $\pm 5\%$ break-even requires sustained position-level capture of ≈ 6.2 – $7.5\times$ the pool average for 90 days. The uncrowded theoretical ceiling for a $\pm 5\%$ band is $41.5\times$ — but it assumes every competing dollar is full-range; on a major pool where liquidity is predominantly concentrated, realized multipliers compress toward $\approx 1\times$, and a sustained 6 – $7\times$ means near-zero competition at the tick for a quarter. Static $\pm 5\%$ on B' requires an in-range-equivalent 1.8% APR $\approx 0.10\times$ the measured pool average — adequacy fails only if pool fee yield collapses by more than $\approx 90\%$ and stays there for the quarter. Positions whose adequacy depends on stress volume holding up are treated as marginal.

Operating rule

(1) Never re-center on range exit at widths inside the measured oscillation amplitude — in either regime; v3 shows this is structural, not empirical luck. (2) In measured oscillatory regimes, hold a static tight range: fee-dense and nearly free (B': 1.2% break-even vs 17.5% measured pool APY). (3) In trending regimes, widen, go full-range, or exit — in asset terms, full-range LP outperformed holding the asset by 10.8% through the drawdown. (4) Re-center on regime change, not on price exit, using the path-efficiency gauge comparatively (A vs B': 0.120 vs 0.004 on daily returns — well over an order of magnitude of separation; absolute thresholds require per-frequency, per-window calibration, a finding this study reports rather than hides). (5) An exit-triggered “self-balancing” automation is a systematic executor of the worst trade in this study — and its trigger levels are publicly predictable on-chain, adding adversarial execution (MEV) as a structural cost of the policy itself, not an implementation detail.

Worst events, read by hand (threshold $\pm 5\%$, window A)

Hour	Price	Value before	Value swapped	Event cost	Trigger
66	\$2,306	\$101,235	\$50,617	\$53.62	exit up
118	\$2,354	\$98,662	\$49,331	\$52.33	exit up
84	\$2,196	\$97,511	\$48,755	\$51.76	exit down
601	\$2,351	\$96,049	\$48,024	\$51.02	exit up
498	\$2,238	\$94,928	\$47,464	\$50.46	exit down

Each of the five largest re-centers swaps roughly half the position at the price extreme that forced the exit; window B' shows the identical signature. Aggregates are where analysis starts, not where it ends.

Stated limitations and production path

This study (decision layer)	Production extension
Fee income inverted (break-even APR); pool-level anchor measured (DeFiLlama; Dune query 7923963 published), position-level accrual not yet measured	Measured position-level accrual from pool events / subgraph
Hourly closes; exits at close; Coinbase price as pool proxy. Bias direction: intra-hour wicks would add triggers and worsen fills for threshold policies — the study understates their damage (conclusion-conservative)	Tick-level simulation on pool data; wick-accurate exits
Two disjoint real windows, one pair; interval estimates from 400 synthetic paths, not yet from rolling real windows	Multi-pair, rolling windows, bootstrap confidence intervals
Regime gauge used comparatively; thresholds not universal	Per-frequency, per-window threshold calibration
Static costs (5+5 bps, fixed gas); MEV named as structural cost	Depth-aware slippage; private execution for any re-center

This package ships the engine, the nine-test validation suite, the reproduction gate, all run scripts, archived inputs, full run outputs and the research log — including the pre-registered predictions that failed and the belief updates they forced, and a v2 erratum. Prior published work: UMA oracle base-rate study (1.17M+ assertions) and related on-chain research under the handle Xpertknight (Dune). Research and decision support; not investment advice; no live trading recommendation.